Lahore University of Management Sciences
Syed Babar Ali School of Science and Engineering
AI-501
Fall 2025
Mid Exam – Part I

Time Allowed: 1 hour **Total Points**: 90 (Weightage: 40%)

Name: _____

Campus ID: _____

Problem 1. Select ALL (up to three) correct choices. Every incorrect answer would earn a penalty of 1 point but the total marks of any multiple-choice question will not be less than zero.

(1) For vectors x and y in $\mathbb{R}^n$, which imply they are orthogonal?
- (a) $x^T y = 0$
- (b) The projection of x on y is zero
- (c) They are linearly dependent
- (d) The angle between them is $90°$

(2) If a vector v is scaled by a constant k, which statements are true?
- (a) The direction remains unchanged for $k > 0$
- (b) The magnitude becomes $|k|\|v\|$
- (c) The vector becomes orthogonal to its original self
- (d) The magnitude remains unchanged for all k

(3) Which of the following statements about norms are true?
- (a) $\|a\| = \sqrt{a^T a}$
- (b) $\|a + b\| = \|a\| + \|b\|$
- (c) $\|ka\| = |k|\|a\|$
- (d) $\|a\| = 0 \Rightarrow a = 0$

(4) Which of the following can represent a unit vector?
- (a) $[1, 0, 0]$
- (b) $[1, 1, 1]$
- (c) $[0.6, 0.8]$
- (d) $[3, 4]/5$

(5) Which of the following are equivalent characterizations of an orthogonal matrix Q?
- (a) $Q^\top Q = I$
- (b) $Q^{-1} = Q^\top$
- (c) Q has all eigenvalues strictly positive
- (d) $\|Qx\|_2 = \|x\|_2$ for all x

(6) Suppose A is 2×2 with $\det(A) \neq 0$. Which are correct?

(a) A is invertible

(b) A must be symmetric

(c) $\det(A^{-1}) = 1/\det(A)$

(d) $\det(cA) = c^2 \det(A)$ for scalar c

(7) Which of the following statements define linear independence correctly?

(a) A set $\{v_1, \ldots, v_k\}$ is linearly independent if $c_1 v_1 + \cdots + c_k v_k = 0$ implies all $c_i = 0$

(b) $v_1, \ldots, v_k$ are independent if none of them are zero vectors

(c) Linear dependence means at least one vector can be written as a linear combination of others

(d) $\det(A) = 0$ indicates independence of columns of A

(8) Which of the following describe the span of a set of vectors?

(a) The set of all linear combinations of given vectors

(b) A subspace containing all those vectors

(c) A single point representing the sum of all vectors

(d) The smallest subspace that contains the given vectors

(9) Let $\mathbf{A}$ be a 3×3 matrix with singular values $\sigma_1 = 6$, $\sigma_2 = 3$, and $\sigma_3 = 2$. What is the **determinant** of $\mathbf{A}^T\mathbf{A}$?

(a) 36

(b) 72

(c) 1296

(d) 216

(10) Suppose $\{u_1, u_2, u_3\}$ are dependent. Which must be true?

(a) At least one vector is a linear combination of the others

(b) $\det([u_1\ u_2\ u_3]) = 0$ if square

(c) All coefficients in any dependence relation are nonzero

(d) $\text{Rank}([u_1\ u_2\ u_3]) = 3$

(11) Which statements about rank and nullity are true for $A \in \mathbb{R}^{m \times n}$?

(a) $\text{rank}(A) + \text{nullity}(A) = n$

(b) $\text{rank}(A) = \text{rank}(A^\top)$

(c) $\text{rank}(A)$ equals number of rows

(d) If $\text{rank}(A) = m$, columns of A are independent

(12) Suppose A is 4×3 and $\text{rank}(A) = 3$. Which are true?

(a) Columns of A are linearly independent

(b) $\text{Null}(A) = \{0\}$

(c) Row space has dimension 3

(d) $\text{Col}(A)$ has dimension 3

(13) A matrix A is invertible with eigenvalues $\lambda_1 = 1$ and $\lambda_2 = 6$. What are the eigenvalues of $4A^3 + 6A^{-1} + 2I$?

(a) $\lambda_1 = 10,\ \lambda_2 = 865$

(b) $\lambda_1 = 12,\ \lambda_2 = 867$

(c) $\lambda_1 = 8,\ \lambda_2 = 860$

(d) $\lambda_1 = 14,\ \lambda_2 = 870$

(14) Let $A = \begin{bmatrix} 2 & 0 \\ 0 & -1 \end{bmatrix}$. Which are true about its associated transformation $T(x) = Ax$?

(a) T stretches x-axis by factor 2

(b) T reflects across x-axis due to -1 on diagonal

(c) $\det(A) = 0$

(d) T is invertible since $\det(A) \neq 0$

(15) For symmetric matrices $A = A^\top$, which are true?

(a) All eigenvalues are real

(b) Eigenvectors for distinct eigenvalues are orthogonal

(c) A cannot be diagonalized by an orthogonal matrix

(d) $A = Q\Lambda Q^\top$ with Q orthonormal

(16) Which are correct regarding determinant/trace and eigenvalues?

(a) $\det(A) = \prod_i \lambda_i$ (over algebraic multiplicity)

(b) $\mathrm{tr}(A) = \prod_i \lambda_i$

(c) If any eigenvalue is zero, then $\det(A) = 0$

(d) If eigenvalues are all nonzero, then A is invertible

(17) In supervised learning, which components are essential?

(a) A labeled dataset (x_i, y_i)

(b) A hypothesis class or model family

(c) An unlabeled test set only

(d) A loss function to evaluate predictions

(18) Which statements about evaluation metrics are true?

(a) Recall measures correctness among predicted positives

(b) Precision measures coverage of actual positives

(c) Accuracy is always reliable under severe class imbalance

(d) F1 is the harmonic mean of precision and recall

(19) In kNN algorithm,

(a) Large k tends to lower bias and higher variance

(b) Large k tends to higher bias and lower variance

(c) k has no effect on model smoothness

(d) Optimal k is often selected via cross-validation

(20) About polynomial regression with degree d:

(a) It is linear in parameters after feature mapping $x \mapsto [1, x, \ldots, x^d]$

(b) Increasing d always improves test error

(c) High d can cause overfitting

(d) Feature scaling can improve numerical stability of the fit

(21) For Ridge regression with $\lambda > 0$:

(a) $\hat{\theta}_\lambda = (X^\top X + \lambda I)^{-1} X^\top y$

(b) As λ increases, coefficients shrink toward 0

(c) Ridge sets some coefficients exactly to zero by design

(d) Ridge increases variance at the cost of lower bias

(22) Best rank-k approximation of matrix A is given by (Eckart–Young):

(a) $A_k = \sum_{i=1}^{k} \sigma_i u_i v_i^\top$ minimizes $\|A - A_k\|_F$

(b) $\|A - A_k\|_F^2 = \sum_{i>k} \sigma_i^2$

(c) Using smaller singular directions preserves of the matrix information in the approximation

(d) Keeping more components always increases reconstruction error

(23) We can reduce dimensionality of the Data using PCA by:

(a) Retaining the first k principal components with largest eigenvalues

(b) Discarding small components reduces noise and redundancy

(c) Retaining all components changes variance distribution

(d) Projection onto top k PCs minimizes reconstruction error in ℓ_2 sense

(24) Consider a classifier for the detection of cancer disease. True here refers to Cancer positive. We do not want to diagnose any cancer positive patient as healthy. Which of the following situations would you prefer for your classifier?

(a) FN $\gg$ FP

(b) FP $\gg$ FN

(c) FN = FP$\times$ TP

(d) TN $\gg$ FP

(25) What is the Frobenius norm of the following matrix A?

$$A = \begin{bmatrix} 0 & 5 \\ 2 & 3 \end{bmatrix}$$

(a) $\sqrt{35}$

(b) $\sqrt{40}$

(c) $\sqrt{38}$

(d) $\sqrt{42}$